

TECHNICAL GUIDE

# Creasing matrix and ejection rubber

Every colour-coded size we extrude, the profiles they come on, how to work out what to trial, and the six steps we would tell any new operator.

Extruded on the Isle of Man since 2009

THE COLOUR IS THE HEIGHT

|                                                                                     |                                                                                      |        |         |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|---------|
|    |    | Pink   | 0.30 mm |
|    |    | Blue   | 0.40 mm |
|    |    | White  | 0.45 mm |
|   |   | Red    | 0.50 mm |
|  |  | Silver | 0.55 mm |
|  |  | Yellow | 0.60 mm |
|  |  | Mauve  | 0.65 mm |
|  |  | Orange | 0.70 mm |
|  |  | Green  | 0.80 mm |
|  |  | Sky    | 1.00 mm |

Drawn to scale — Phoenix+, 10mm standard centre. Each height has its own colour, so an operator reads the size off the strip from across the room. Heights are exaggerated against width, or a 0.30mm shoulder on a 10mm base would be a hairline.

WHO WE ARE

The complaints came to us

Before 2009 we supplied creasing matrix to this industry, which meant we were the ones who took the call when it lifted, when it crushed, when it left residue on the plate. Converters and die-makers told us plainly, year after year, that it was not good enough.

They were right. And we knew exactly why, because we knew what was going into it. So in 2009 we built the line ourselves, on the Isle of Man, and specified the polymer and the adhesive ourselves rather than accepting what a supplier offered. Everything that leaves here is extruded by us. Nothing is bought in and relabelled, which means every tolerance is ours to hold and ours to answer for.

THE RANGE

Four ranges. Not forty.



Phoenix+

Our standard synthetic matrix for folding carton, on 7mm, 10mm and 12mm bases. The one most customers start with.

|          |                |
|----------|----------------|
| Board    | Folding carton |
| Bases    | 7, 10 and 12mm |
| Material | Virgin polymer |



Phoenix XL

A 12mm or 15mm base for heavy solid board and micro-flute, where a narrower footprint would shift under pressure.

|       |                       |
|-------|-----------------------|
| Board | Heavy solid and flute |
| Bases | 12 and 15mm           |
| Grip  | No lead-edge taping   |



Ultra-SR

A direct swap for steel-based matrix on abrasive recycled board, at the same thickness – without the rust or the perforation marks.

|          |                         |
|----------|-------------------------|
| Board    | Recycled and greyboard  |
| Material | High impact polyester   |
| Benefit  | No rust, no perforation |



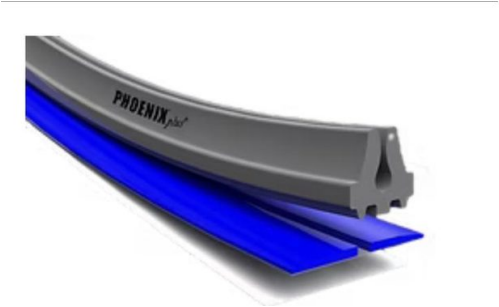
Exceed Rubber

Ejection rubber that holds its profile past 1.6 million impressions instead of going hard and losing rebound.

|          |                        |
|----------|------------------------|
| Run life | 1,600,000+ impressions |
| Benefit  | Reduced angel hair     |
| Widths   | 5.00 and 7.50mm        |

PHOENIX+

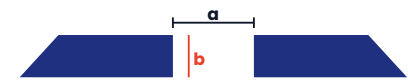
# Our standard matrix, and the one most presses run



Phoenix+ is where our polymer and adhesive work ends up. It is built for long, high-speed runs on folding carton, and it is the range we hold most sizes of on the shelf.

|                                                                                                             |                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Clean-peel tape</b><br>Holds under pressure, lifts without residue, no cleaning the platen between jobs. | <b>Self-locating bridges</b><br>Small enough for tight areas and thermally bonded, so nothing is left behind in the channel. |
| <b>Virgin polymer</b><br>Not regrind. It is what gives the channel walls their run life.                    | <b>Smooth shoulder</b><br>Clean sheet transfer without reworking the shoulder by hand.                                       |

## 7mm mini centre



| BASE COLOUR                                                                                | HEIGHT  | CHANNEL WIDTHS (MM)                                                  |
|--------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------|
|  Pink   | 0.30 mm | 0.5, 0.6, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9 |
|  Blue   | 0.40 mm | 0.5, 0.6, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9 |
|  White  | 0.45 mm | 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9           |
|  Red    | 0.50 mm | 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.9, 2.1, 2.3      |
|  Silver | 0.55 mm | 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.9, 2.1, 2.3      |
|  Yellow | 0.60 mm | 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.9, 2.1                               |
|  Orange | 0.70 mm | 1.5, 1.6, 1.7, 1.9, 2.1                                              |

28 strips × 750mm — 21 metres per box

## 7mm off-centre



| BASE COLOUR                                                                              | HEIGHT  | CHANNEL WIDTHS (MM)                                                  |
|------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------|
|  Pink   | 0.30 mm | 0.5, 0.6, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9 |
|  Blue   | 0.40 mm | 0.5, 0.6, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9 |
|  White  | 0.45 mm | 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9           |
|  Red    | 0.50 mm | 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.9, 2.1, 2.3      |
|  Silver | 0.55 mm | 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.9, 2.1, 2.3      |
|  Yellow | 0.60 mm | 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.9, 2.1, 2.3, 2.5                     |
|  Mauve  | 0.65 mm | 1.5, 1.6, 1.7, 1.9, 2.1, 2.3, 2.5                                    |
|  Orange | 0.70 mm | 1.5, 1.6, 1.7, 1.9, 2.1, 2.3, 2.5                                    |

28 strips × 750mm — 21 metres per box

## 10mm standard centre



| BASE COLOUR                                                                                | HEIGHT  | CHANNEL WIDTHS (MM)                                                  |
|--------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------|
|  Pink   | 0.30 mm | 0.5, 0.6, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9 |
|  Blue   | 0.40 mm | 0.5, 0.6, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9 |
|  White  | 0.45 mm | 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9           |
|  Red    | 0.50 mm | 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.9, 2.1, 2.3      |
|  Silver | 0.55 mm | 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.9, 2.1, 2.3      |
|  Yellow | 0.60 mm | 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.9, 2.1, 2.3, 2.5, 2.7, 2.8, 3.0, 3.2 |
|  Mauve  | 0.65 mm | 1.5, 1.6, 1.7, 1.9, 2.1, 2.3, 2.5, 2.7, 2.8, 3.0, 3.2                |
|  Orange | 0.70 mm | 1.5, 1.6, 1.7, 1.9, 2.1, 2.3, 2.5, 2.7, 2.8, 3.0, 3.2                |
|  Green  | 0.80 mm | 1.5, 1.6, 1.7, 1.9, 2.1, 2.3, 2.5, 2.7, 2.8, 3.0, 3.2                |
|  Sky    | 1.00 mm | 2.3, 2.5, 2.7, 2.8, 3.0, 3.2                                         |

32 strips × 750mm — 24 metres per box

## 12mm centre



| BASE COLOUR                                                                              | HEIGHT  | CHANNEL WIDTHS (MM)    |
|------------------------------------------------------------------------------------------|---------|------------------------|
|  Yellow | 0.60 mm | 3.5*, 4.0#, 4.5#       |
|  Mauve  | 0.65 mm | 3.5*, 4.0#, 4.5#       |
|  Orange | 0.70 mm | 3.5*, 4.0#, 4.5#, 5.0# |
|  Green  | 0.80 mm | 3.5*, 4.0#, 4.5#, 5.0# |
|  Sky    | 1.00 mm | 3.5*, 4.0#, 4.5#, 5.0# |
|  Silver | 1.20 mm | 3.5*, 4.0#, 4.5#, 5.0# |
|  Lime   | 1.50 mm | 3.5*, 4.0#, 4.5#, 5.0# |

\* 3.5mm: 27 strips × 750mm — 20.25 metres

# 4.0–5.0mm: 20 strips × 750mm — 15 metres

## 12mm double crease



| BASE COLOUR                                                                               | HEIGHT  | CHANNEL WIDTHS (MM) |
|-------------------------------------------------------------------------------------------|---------|---------------------|
|  Blue  | 0.40 mm | 1.0, 1.3, 1.5       |
|  White | 0.45 mm | 1.0, 1.3, 1.5       |
|  Red   | 0.50 mm | 1.0, 1.3, 1.5       |

24 strips × 750mm — 18 metres per box

PHOENIX XL

# A wider footprint for board that fights back



On heavy solid board and micro-flute, a narrow base can shift under pressure. Phoenix XL puts the same polymer on a 12mm or 15mm flange so it stays where you put it.

## No lead-edge taping

Enough platen grip out of the box that you do not need to tape the leading edge.

## 12mm and 15mm bases

Pick the footprint to suit the caliper and the pressure you are running.

## High-impact polymer

Resists channel wall collapse across multi-million impression runs.

## 12mm standard centre



| BASE COLOUR                                                                                | HEIGHT  | CHANNEL WIDTHS (MM)                              |
|--------------------------------------------------------------------------------------------|---------|--------------------------------------------------|
|  Blue   | 0.40 mm | 1.3, 1.4, 1.5, 1.6, 1.7                          |
|  Red    | 0.50 mm | 1.3, 1.4, 1.5, 1.6, 1.7, 1.9                     |
|  Yellow | 0.60 mm | 1.5, 1.6, 1.7, 1.9, 2.1, 2.3, 2.5, 2.7, 3.0, 3.2 |
|  Orange | 0.70 mm | 2.1, 2.3, 2.5, 2.7, 3.0, 3.2                     |
|  Green  | 0.80 mm | 2.1, 2.3, 2.5, 2.7, 3.0, 3.2                     |
|  Sky    | 1.00 mm | 2.1, 2.3, 2.5, 2.7, 3.0, 3.2                     |

32 strips × 750mm — 24 metres per box

## 15mm extra-wide centre



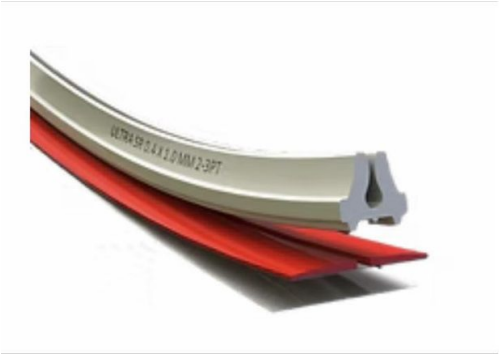
| BASE COLOUR                                                                                | HEIGHT  | CHANNEL WIDTHS (MM)    |
|--------------------------------------------------------------------------------------------|---------|------------------------|
|  Yellow | 0.60 mm | 3.5*, 4.0#, 4.5#, 5.0# |
|  Orange | 0.70 mm | 3.5*, 4.0#, 4.5#, 5.0# |
|  Green  | 0.80 mm | 3.5*, 4.0#, 4.5#, 5.0# |
|  Sky    | 1.00 mm | 3.5*, 4.0#, 4.5#, 5.0# |

\* 3.5mm: 27 strips × 750mm — 20.25 metres

# 4.0–5.0mm: 20 strips × 750mm — 15 metres

ULTRA-SR

# A steel replacement that cannot rust



Ultra-SR is a high-impact polyester base at the same thickness as steel matrix, so it drops straight into a steel specification. On abrasive recycled board it holds crease depth without rusting or leaving perforation marks.

|                                                                                                    |                                                                                                     |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Formulated adhesive</b><br>Specified for maximum impressions rather than for easy manufacture.  | <b>Steel compatible</b><br>Same thickness as steel matrix, so existing specifications carry across. |
| <b>No rust, no perforation</b><br>The two things that end a steel matrix's life on recycled board. | <b>Fingerlift tape as standard</b><br>On every size, for faster makeready.                          |

## 7mm centre



| BASE COLOUR | HEIGHT  | CHANNEL WIDTH | RULE  |
|-------------|---------|---------------|-------|
| Mauve       | 0.38 mm | 0.5 mm        | 2-3pt |
| Silver      | 0.38 mm | 0.8 mm        | 2-3pt |
| Buff        | 0.38 mm | 1.0 mm        | 2-3pt |
| Violet      | 0.38 mm | 1.2 mm        | 2-3pt |
| White       | 0.45 mm | 1.3 mm        | 2-3pt |
| Sky         | 0.45 mm | 1.4 mm        | 2-3pt |
| Lime        | 0.45 mm | 1.5 mm        | 2-3pt |
| Yellow      | 0.50 mm | 1.6 mm        | 2-3pt |
| Rose        | 0.50 mm | 1.7 mm        | 2-3pt |
| Slate       | 0.55 mm | 1.9 mm        | 2-3pt |
| Green       | 0.60 mm | 2.0 mm        | 2-3pt |
| Brown       | 0.60 mm | 2.1 mm        | 2-3pt |
| Red         | 0.70 mm | 2.3 mm        | 2-3pt |
| Black       | 0.70 mm | 2.4 mm        | 2-3pt |
| Blue        | 0.80 mm | 2.6 mm        | 2-3pt |
| Orange      | 0.90 mm | 3.0 mm        | 3-4pt |

32 strips × 750mm — 24 metres per box

## 7mm off-centre



| BASE COLOUR                                                                              | HEIGHT  | CHANNEL WIDTH | RULE  |
|------------------------------------------------------------------------------------------|---------|---------------|-------|
|  Silver | 0.38 mm | 0.8 mm        | 2-3pt |
|  Buff   | 0.38 mm | 1.0 mm        | 2-3pt |
|  Violet | 0.38 mm | 1.2 mm        | 2-3pt |
|  White  | 0.45 mm | 1.3 mm        | 2-3pt |
|  Sky    | 0.45 mm | 1.4 mm        | 2-3pt |
|  Lime   | 0.45 mm | 1.5 mm        | 2-3pt |
|  Yellow | 0.50 mm | 1.6 mm        | 2-3pt |
|  Rose   | 0.50 mm | 1.7 mm        | 2-3pt |

32 strips × 750mm — 24 metres per box

## 12mm centre



| BASE COLOUR                                                                               | HEIGHT  | CHANNEL WIDTH | RULE            |
|-------------------------------------------------------------------------------------------|---------|---------------|-----------------|
|  Grey  | 1.30 mm | 3.80 mm       | 3-4pt and 4-6pt |
|  Black | 1.50 mm | 5.00 mm       | 3-4pt and 4-6pt |
|  Cream | 1.70 mm | 6.00 mm       | 3-4pt and 4-6pt |

20 strips × 750mm — 15 metres per box

## 12mm double crease



| BASE COLOUR                                                                                | HEIGHT  | CHANNEL WIDTHS (MM)    |
|--------------------------------------------------------------------------------------------|---------|------------------------|
|  Buff   | 0.38 mm | 1.0mm in D3, D4 and D5 |
|  White  | 0.45 mm | 1.3mm in D3, D4 and D5 |
|  Yellow | 0.50 mm | 1.6mm in D3, D4 and D5 |

24 strips × 750mm — 18 metres per box



## EXCEED RUBBER

# Still the same profile at 1.6 million impressions



Ejection rubber that holds its shape rather than degrading and losing rebound. That means shorter makeready and noticeably less angel hair.

**1.6 million impressions**

No measurable drop in performance across the run.

**UV resistant**

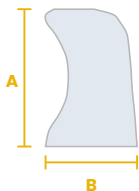
Does not degrade under shop floor lighting between jobs.

**Heat resistant**

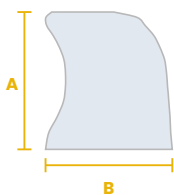
Stays flexible instead of going brittle under press friction.

**Dimensionally stable**

Sits down on the die board accurately first time.

**Exceed 5.00mm**

| PRODUCT                | HEIGHT | WIDTH  | COLOUR | PER BOX                |
|------------------------|--------|--------|--------|------------------------|
| Exceed 7.00mm × 5.00mm | 7.00mm | 5.00mm | White  | 24 metres (32 × 750mm) |
| Exceed 7.00mm × 5.00mm | 7.00mm | 5.00mm | White  | 60 metres (80 × 750mm) |
| Exceed 7.25mm × 5.00mm | 7.25mm | 5.00mm | White  | 24 metres (32 × 750mm) |
| Exceed 7.25mm × 5.00mm | 7.25mm | 5.00mm | White  | 60 metres (80 × 750mm) |

**Exceed 7.50mm**

| PRODUCT                | HEIGHT | WIDTH  | COLOUR | PER BOX                |
|------------------------|--------|--------|--------|------------------------|
| Exceed 7.00mm × 7.50mm | 7.00mm | 7.50mm | White  | 24 metres (32 × 750mm) |
| Exceed 7.00mm × 7.50mm | 7.00mm | 7.50mm | White  | 60 metres (80 × 750mm) |
| Exceed 7.25mm × 7.50mm | 7.25mm | 7.50mm | White  | 24 metres (32 × 750mm) |
| Exceed 7.25mm × 7.50mm | 7.25mm | 7.50mm | White  | 60 metres (80 × 750mm) |

SIZING

Working out a starting size

Depth is roughly the caliper of the board. Width is the caliper multiplied by one and a half, plus the thickness of the creasing rule. Round to the nearest size we make, and treat the result as the size to trial first — board behaves differently press to press.

Depth ≈ caliper

Width = (caliper × 1.5) + rule

The size finder at [creasingmatrix.com](https://creasingmatrix.com) does this for you, and then lists every size we actually extrude that fits — by colour, range and profile.

ON THE BENCH

Fitting matrix in six steps

|                                                                                                                                   |                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>1 Clean the plate</b><br>Degrease and dry the cutting plate or counter jacket. Anything left on it will show up in the crease. | <b>2 Cut to length</b><br>Use a proper matrix cutter at 45° or 90°. A clean end sits down flat.           |
| <b>3 Locate on the rule</b><br>Push the locator bridge down over the creasing rule so the channel sits square.                    | <b>4 Peel the backing</b><br>Lift the fingerlift tape to expose the adhesive, keeping the strip in place. |
| <b>5 Take an impression</b><br>One cycle at normal pressure transfers the matrix onto the plate.                                  | <b>6 Remove the bridges</b><br>Lift the locators out of the channel and you're ready to run.              |

ACCESSORIES

Shim and patching tape

SELF-ADHESIVE SHIM TAPE

Super Shim Plus

8.0mm wide on a 12mm fingerlift backing, with an adhesive specified for this job specifically — it holds long term rather than creeping under pressure.

Made from hardened stainless steel, so it won't compress in use.

| THICKNESS            | ROLL                    |
|----------------------|-------------------------|
| 0.03mm — 30 microns  | 8mm wide, 10 metre roll |
| 0.05mm — 50 microns  | 8mm wide, 10 metre roll |
| 0.10mm — 100 microns | 8mm wide, 10 metre roll |

PATCHING TAPE

Profitape

Colour-coded self-adhesive patching tape for precise cutting and foil embossing, translucent enough to see your lines through.

| THICKNESS | ROLL                  |
|-----------|-----------------------|
| Blue      | 3mm × 30 micron × 35m |
| Blue      | 6mm × 30 micron × 35m |
| Red       | 3mm × 50 micron × 25m |
| Red       | 6mm × 50 micron × 25m |
| Yellow    | 3mm × 80 micron × 18m |
| Yellow    | 6mm × 80 micron × 18m |

## TROUBLESHOOTING

# Nine things that go wrong, and what causes them

Most of these are not the matrix. We would rather tell you that than sell you something that will not fix it. Each one has its causes and what to change at [creasingmatrix.com/#faults](https://creasingmatrix.com/#faults).

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### Crease cracks when it is folded

The crease looks clean coming off the press and splits on the folder-gluer.

### The crease bursts on the press

Fibres split along the crease as it is cut, not later.

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### The matrix lifts off the plate

It moves during the run, or comes away with the sheet.

### Residue left behind at changeover

The matrix comes up but the adhesive stays on the plate.

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### The matrix crushes or loses its profile

Creases go shallow part-way through the run and the shoulders spread.

### The crease is off-centre

One shoulder is deeper than the other, and the carton folds skew.

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### Splitting at the corners of a crease

The crease is sound along its length and fails where it meets another.

### Angel hair and dusting

Fine fibres around the cut edge, all over the sheet and into the machine.

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### Ejection rubber goes hard

It worked when it went on and now it does not push the waste out.

## GET IN TOUCH

# Talk to us

Describe what you are seeing and there is someone on the other end who has stood at a press with the same problem.

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[creasingmatrix.com](https://creasingmatrix.com)

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